

Principles of Sequence Alignment

14/06/2026

Phuc-Loi Luu, PhD

Global vs Local Alignment

Local Alignment

Target Sequence

5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'

Query Sequence

5' TACTCACGGATGAGGTACTTTAGAGGC 3'

||||| ||||| ||||| ||||| |||||

Global Alignment

Target Sequence

5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'

||||| ||||| ||||| ||||| |||||

5' ACTACTAGATT---ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'

Query Sequence

input
string

HEAGAWGHEEAHGEGAE
PAWHEAEHE

Global alignment

HEAGAWGHEEAHGEGAE
--|-|-|-|-|-|-|-|
--P-AW-H-EA--E-HE

Local alignment

AWGHEEAH
|-|||||
AW-HEAEH

1. Global Alignment (Needleman-Wunsch)

- Global alignment attempts to align every character in every sequence from the beginning to the end.
- It is best used when two sequences are of similar length and are expected to be homologous across their entire span.
 - What: Aligning two full-length sequences (e.g., comparing two versions of the same gene from different species).
 - How: It uses the Needleman-Wunsch algorithm, a dynamic programming approach that maximizes the total alignment score across the entire length of both sequences, including gaps at the ends if necessary.
 - Why: To determine the total evolutionary distance or structural conservation between two related sequences.
- Example: Comparing the entire nucleotide sequence of the human insulin gene to that of a chimpanzee to identify overall conservation rates.

Local Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'

Query Sequence 5' TACTCACGGATGAGGTACTTTAGAGGC 3'

|||| ||||| |||||

Global Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'

||||| ||||| ||||| |||||

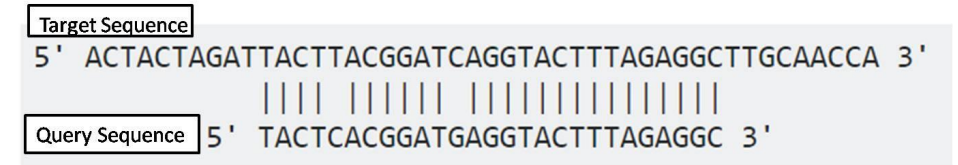
Query Sequence 5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'

2. Local Alignment (Smith-Waterman)

- Local alignment searches for the most similar sub-regions (motifs or domains) within larger sequences.
- It ignores the parts of the sequences that do not match well.
 - What: Finding conserved functional domains, gene families, or short regulatory motifs hidden within large, otherwise dissimilar sequences.
 - How: It uses the Smith-Waterman algorithm. Unlike the global approach, this algorithm assigns a score of zero to any negative alignment segments, effectively "restarting" the alignment whenever a match is found and ending it when the similarity drops off.
 - Why: To identify functional modules, binding sites, or domains that have been conserved through evolution even when the rest of the genetic context has diverged or been rearranged.
- Example: Searching for a specific protein domain (like a zinc finger motif) within a large, uncharacterized protein sequence. BLAST (Basic Local Alignment Search Tool) is the most common implementation of this.

Local Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
Query Sequence 5' TACTCACGGATGAGGTACTTTAGAGGC 3'



Global Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
Query Sequence 5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'



Comparison Summary

Feature	Global Alignment	Local Alignment
Objective	Full sequence similarity	Best matching sub-segments
Algorithm	Needleman-Wunsch	Smith-Waterman
Best Used When	Sequences are similar in length	Sequences are different or large
Primary Use	Evolutionary relationship	Motif/Domain identification

Practical Guidance: When to choose which?

- **Use Global Alignment** if you are performing a comparative genomics study on well-defined, orthologous genes where you expect the sequences to be roughly the same size and essentially "the same" thing.
- **Use Local Alignment** for database searching (like BLAST). If you have a sequence "read" from a sequencer and you want to know what it is, you don't care about the whole sequence of every gene in the database; you want to find the specific "home" for that read.

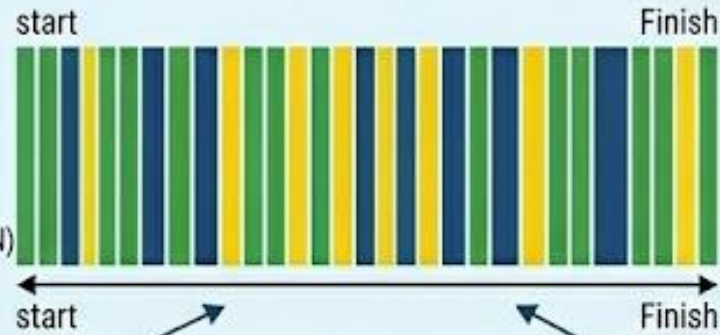
GENOMIC SEQUENCE ALIGNMENT STRATEGIES

GLOBAL ALIGNMENT (Needleman-Wunsch)

Alignment of whole-length homologous protein sequences

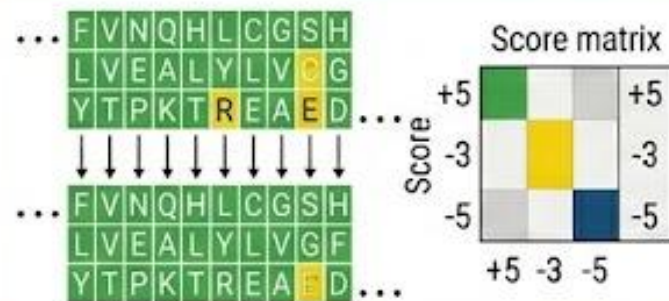
Top Sequence 1
(e.g., HUMAN INSULIN)

Bottom Sequence 2
(e.g., CHIMPANZEE INSULIN)
110 aa



Example:
Human Insulin A
(110 aa)

Chimpanzee Insulin A
(110 aa)



LOCAL ALIGNMENT (Smith-Waterman/BLAST)

Finding a **specific domain** within large, dissimilar sequences

Top Sequence 3
(e.g., Large Unknown Protein
1500 aa)

Bottom Sequence 4
(e.g., *Drosophila* Zn-Finger
60 aa, 100% ID with A)



Example:

- Match
- Mismatch
- Mismatch
- Gap
- Domain region

